

Population decoding of visual motion direction.

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Introduction

Studying the **internal representation** of natural features in the primary visual cortex (V1), like direction or spatial frequency, is crucial to understand how we perceive the external world. Research on 2D motion direction in **non-human primates** in particular when displaying **naturalistic stimuli** like MotionClouds reveals substantial diversity and multiple mechanisms within the neuronal population. The aim of this project is to examine how a large population of V1 neurons encodes stimulus direction and how this representation is modulated by the uncertainty using classical and decoding methods.

Methods

Neuronal recordings were made in V1 of one marmoset monkey using the Neuropixel technology during 2D motion presentation in twelve directions (θ) from 0° to 375° , four levels of orientation bandwidth (B_θ ; 0° , 15° , 30° and 45°) and 4 levels of spatial frequency bandwidth (B_{sf} ; 0.01, 0.175, 0.7 and 2.8 cpd) from the MotionClouds python library.

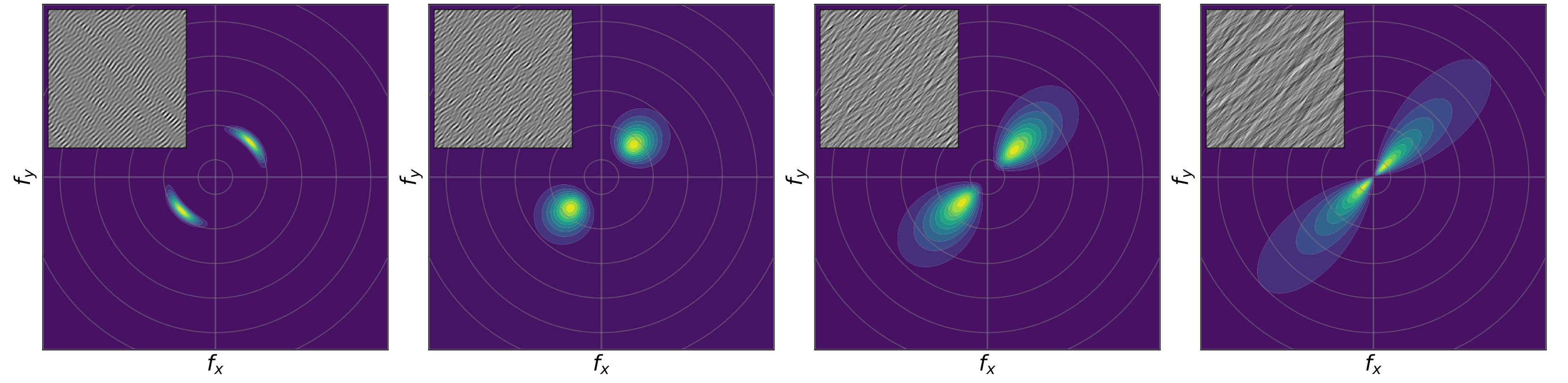


Figure 1: Fourier space representation of MotionClouds stimuli. The image is generated by the inverse fourier transform after the application of an random phase.

The classical tuning curve analysis is based on the optimization of the following equations parameters (κ_ϕ , κ_θ , θ_0 ...) in order to predicted the activity (A) of each neuron in each condition.

$$A_\theta = \kappa_\theta \times (\cos(2 \times (\theta - \theta_0)) - 1)$$

$$A_\phi = \kappa_\phi \times (\cos(\theta - \theta_0) - 1)$$

$$A = \exp(A_\theta + A_\phi + \log R_{amp}) + R_0$$

Based on our extracted parameters, orientation and direction selective indexes (OSI and DSI) are computed as following :

$$DSI = \frac{R_{pref} - R_{null}}{R_{pref}} = \frac{R_{amp}(1 - e^{-2\kappa_\phi})}{R_0 + R_{amp}}$$

$$OSI = \frac{R_{pref} + R_{null} - (R_{orth+} + R_{orth-})}{R_{pref} + R_{null}}$$

$$= \frac{R_{amp} \times (1 + e^{-2\kappa_\phi} - 2e^{-(2\kappa_\theta + \kappa_\phi)})}{2R_0 + R_{amp} \times (1 + e^{-2\kappa_\phi})}$$

The decoding method optimizes the weights of a multinomial logistic regression to achieve optimal decoding accuracy on a training set.